

Hands-on Learning Activity (Lesson 3)

This activity is designed to test and reinforce your knowledge gained in the Lesson 3.

This activity is optional and based on what you learned in the previous two videos you are requested to try-out these steps and share your results in the format with others for peer review.

Overview

In this assignment, you will apply concepts of portfolio optimization using generative AI (GenAI). By leveraging tools like Python, Yahoo Finance, and GenAI-driven methods, you will analyze stock data, calculate portfolio variance, and optimize portfolio weights to minimize risk. You will also explore the efficient frontier to visualize optimal risk-return combinations, gaining insights into data-driven decision-making for portfolio management.

Learning Objectives

By completing this assignment, students will be able to:

- 1. Apply GenAI Models for Market and Financial Analysis:** Utilize generative AI to fetch and analyze historical market data for multiple stocks.
- 2. Design GenAI-Based Trading Strategies to Test Trading Hypotheses:** Leverage GenAI tools to calculate portfolio variance and covariance matrices for risk assessment.

3. Utilize GenAI for Portfolio Management and Optimization: Optimize portfolio allocations to minimize variance and explore the efficient frontier for optimal portfolio design.

4. Leverage GenAI Tools for Competitive Advantage in Finance: Gain hands-on expertise in GenAI-powered tools for advanced financial analysis, enabling data-driven portfolio management and research.

Project Steps

1. Set Up Your Python Environment:

Install all required libraries, such as Pandas, NumPy, Matplotlib, Scipy, and Yfinance.

2. Gather and Explore Data:

Use GenAI to fetch historical price data for a selection of 5-10 stocks using **Yahoo Finance**.

Analyze the data to calculate **daily returns** and **average returns** for each stock.

3. Understand Portfolio Variance:

Ask GenAI for a brief explanation of **portfolio variance** and its importance in creating a minimum variance portfolio.

Use GenAI to get the Python code for calculating the **covariance matrix** and **portfolio variance**.

4. Optimize the Portfolio:

→ Use **optimization techniques** (e.g., `scipy.optimize`) to determine the weights of the portfolio that minimize variance.

→ Include a constraint that the sum of portfolio weights equals 1.

→ Add a target for minimum acceptable returns and solve for the **efficient frontier**.

5. Visualize and Analyze:

- Generate a plot of the **efficient frontier** using Python.
- Highlight the point representing the minimum variance portfolio on the graph.

6. Summarize and Share Findings:

→ Prepare a report including:

- The selected stocks and their historical data summary.
- Portfolio weights for the minimum variance portfolio.
- Graphical representation of the efficient frontier.
- Key insights and recommendations from the analysis.

Save your report in PDF or XLSX format.

7. Share and Review:

- Upload your findings to the **peer-review section**.
- Review and provide feedback on the submissions of at least two fellow learners.